E^2 DNet: An Ensembling Deep Neural Network for Solving Nonconvex Economic Dispatch in Smart Grid

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***Abstract*—Currently, a nonconvex economic dispatch problem is one of the research focuses in the field of smart grid (SG). A variety of algorithms are developed to solve it. However, these algorithms are prone to suffering from high computation cost and slow convergence rate, which creates an inevitable gap between theoretical analysis and practical real-time operations. In this article, we aim at providing an ensemble deep-learning-based approach to tackle such a challenging issue. First, a novel ensemble method is presented to explore the ground truth of nonconvex economic dispatch problems. Second, considering the time-varying total load demand, cost coefficients, and dispatchability of all generation units in a practical SG system as the features, a new deep neural network structure is proposed to learn the complex mapping from instant features to an optimal nonconvex economic dispatch solution. If such a mapping is well approximated by the designed deep neural network, no significant effort is required to solve a new economic dispatch problem, and the solution is obtained on the scale of milliseconds. Third, analyzing that a single deep neural network may be weak to a small part of the mapping space of the nonconvex economic dispatch problem, we further present an ensemble of multiple parallel deep neural networks trained sequentially with a simplified Adaboost.R2 algorithm. Finally, case studies reveal that the proposed approach achieves orders of magnitude speedup in computational time while guaranteeing similar or better performance on minimizing the overall generation cost compared to the state-of-the-art nonconvex economic dispatch algorithms.**

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***Index Terms*—Deep neural network (DNN), economic dispatch (ED), ensemble learning, learning to optimize, nonconvex optimization.**

I. Introduction

IN RECENT years, a smart grid (SG) has proved to be a universal topic for researchers with its tremendous prospects for development. Different from traditional power systems powered by fossil-fueled thermal power generators, SG systems may include more renewable energy sources carrying a time-varying and intermittent power output, which further puts forward stricter accuracy and real-time requirements for its control technology. As a fundamental but challenging optimization problem in SGs, nonconvex economic dispatch (ED) has been studied since the 1990s [1]. In fact, heuristic algorithms have played a vital role in solving nonconvex ED problems, while the majority of them are based on the *population* optimization mechanism and operated in an *iterative* way, resulting in considerable slow convergence speed and high computational cost [2]. Therefore, despite the fact that outstanding solution performance of these heuristic algorithms has been shown through a large number of simulation experiments on different test systems, there is still room for improvement before applying them in practical power systems.

For decades, a deep learning technique has achieved a tremendous success in image recognition [3], natural language processing [4], and many other areas [5] due to its excellent representation capability of complex mapping relationship and simple forward propagation operation [6]. Inspired by this realization, an interesting idea springs up in our minds: *Can we treat the mapping space of a nonconvex ED problem from the system parameters to the optimal solutions as a "black box" and then employ a deep neural network (DNN) to approximate it?* If such a learning task can be well completed, then we must be willing to use simple matrix multiplication calculations in the DNN to replace complicated time-consuming numerical/heuristic operations to achieve real-time nonconvex ED.

It is worth mentioning that there have been some pioneering works applying DNN for ED problems, which can be traced back to the 1990s [7]. However, the computational resources at that time are very limited, resulting in only a single hidden layer designed to be included in the forward neural network, which prevents such a prospective idea from emerging in full force. Recently, Yang *et al.* [8] have employed the DNN for solving ED problems, where the load demand vector is taken as the feature. Also, Guo *et al.* [9] propose to use distributed DNNs to learn the mapping relationship between the average load demand of the entire power network and the optimal output of each local generator so that the DNNs can provide the ED solution in a distributed manner. In [10], the reason why the DNN can approximate a specific optimization algorithm is further theoretically analyzed based on the "deep unfolding" method. Nevertheless, there are still some issues worthy of further study. First, previous related works [7]–[10] only consider convex ED problems, while nonconvex ones are more general and, thus, valuable as a variety of nonconvex components including multiple fuel options (MFOs), valve-point effects (VPEs), and prohibited operating zones (POZs) are widespread in practical power systems [11]. More importantly, only a single convex optimization algorithm is employed in such DNN-based ED approaches to obtain the ground truth since there can only be one global optimal solution for a convex ED problem, which has been guaranteed to be found by existing convex optimization algorithms. However, for a class of nonconvex ED problems, given different problem parameters (e.g., total load demand, unit status, and cost coefficients), different algorithms often show different performances. In other words, if only a single nonconvex ED algorithm is utilized to explore the ground truth, the trained network may not always provide better performance than other nonconvex ED algorithms. Therefore, how to explore the ground truth of nonconvex ED problems is still a challenging topic. Second, some common time-varying parameters such as time-varying unit status and generation cost parameters have not been considered in existing DNN-based ED approaches, entailing that they may not be able to handle dynamic operating scenarios in SGs. Therefore, how to enable the DNN to deal with such varying conditions in practical SGs is also an important area to be explored. Third, a single DNN may perform unsatisfactorily on a small part of the mapping space of the nonconvex ED problem since the optimizer usually focuses on minimizing the *average* loss between the ground truth and the outputs of the DNN. Thus, how to enhance the performance on minimizing the overall generation cost of a single DNN is still a challenging issue to be addressed.

Motivated by the above concerns, in this article, we first present a framework ensembling multiple state-of-the-art nonconvex ED algorithms to explore the ground truth of nonconvex ED problems. Then, with the obtained ground truth, we propose a DNN named *economic dispatch network* (EDNet), whose network structure is well designed so that it can adapt to varying unit status and cost coefficients and produce the optimal nonconvex ED strategies in real time. Finally, analyzing that a single EDNet may be weak to a small part of the mapping space of the nonconvex ED problem, we further propose a novel ensemble ED framework, named ensemble EDNet (*E^2 DNet*). It consists of several EDNets in parallel generating different near-optimal solutions for the nonconvex ED problem, and the one owning least generation cost is selected as the output of E^2 DNet. In order to overcome the aforementioned shortcoming of a single EDNet more efficiently, we also design a simplified Adaboost.R2 algorithm as the training strategy for E^2 DNet, which increases the weight of "hard" samples so that the subsequently trained EDNet can better approximate the mapping space represented by these samples. On one hand, compared with other state-of-the-art nonconvex ED algorithms, the proposed EDNet and E^2 DNet can achieve orders of magnitude speedup in computational time while guaranteeing similar or better performance on minimizing the overall generation cost. On the other hand, compared with some existing DNN-based ED approaches, the proposed EDNet and E^2 DNet can deal with nonconvex components and varying operating conditions in practical SGs. Also, the comparison results between a single EDNet and E^2 DNet reveal the improvement in terms of performance on minimizing the overall generation cost brought by the introduced ensemble framework. Such comparison results distinguish our work from previous studies.

In summary, the main contributions of this article are given as follows.

1) EDNet, a novel learning-based approach for solving a nonconvex ED problem, is proposed, which achieves similar or better performance on minimizing the overall generation cost to the state-of-the-art nonconvex ED algorithms. Different from some DNN-based approaches for convex ED [7]–[10], a novel ensemble method is proposed to explore the ground truth of nonconvex ED problems, so as to guarantee the optimal performance of the EDNet. To our best knowledge, the DNN has not been used to address the nonconvex ED problem before.
2) The proposed EDNet obtains the near-optimal solution on the scale of *milliseconds*, indicating orders of magnitude computational speedup compared to the state-of-the-art nonconvex ED algorithms. Obviously, it is entirely suitable for the real-time requirements of SGs. Besides, compared with some existing DNN-based ED approaches [7]–[10], the proposed EDNet shows excellent flexibility to varying power load demand, unit status, and generation cost functions.
3) E^2 DNet, an ensemble framework of EDNet, is further proposed, which does not only inherit all the advantages of EDNet but also provides *better* performance on minimizing the overall generation cost.

To promote further study of our ensemble learning-based approach for solving more problems far beyond nonconvex ED, the code used to generate the simulation results in this article is available online.[1]

The rest of this article is organized as follows. In Section II, the nonconvex ED problem is formulated, where the unit status is introduced as a parameter of the SG system together with the total load and the power generation cost function. Section III introduces the design and training details of our proposed EDNet, while Section IV further proposes an ensemble framework of EDNet, i.e., E^2 DNet, based on a simplified Adaboost.R2 algorithm. The real-time and generation cost performances of EDNet and E^2 DNet are validated by several case studies in Section V. Finally, Section VI concludes this article.

[1][Online]. Available: https://github.com/599143868/E-2DNet-TII

II. Problem Formulation

A. ED With Nonfixed Dispatchable Generators

The conventional ED problem in SGs aims to find the optimum power allocation among a group of *fixed* dispatchable generators (DGs) subject to the generation–demand balance constraint and local generation constraints. However, in practical scenarios, not all possibly dispatchable generators (PDGs) can always maintain a "dispatchable" state. For example, due to short-term rental or sudden failures, some PDGs are temporarily nondispatchable. Motivated by this consideration and with the assumption that the system involves POZs and transmission losses (TLs), we introduce the time instants $t \in \mathbb{R}$ and the DG states $U \in \{0, 1\}$ to reformulate the ED problem as follows:

$$\min_{P} \sum_{i \in PDG} f_{i,t}(P_{i,t}) \tag{1a}$$

$$\text{s.t.} \sum_{i \in PDG} P_{i,t} = P_{d,t} + P_{l,t} \tag{1b}$$

$$\begin{aligned} &U_{i,t}P_i^{\min} \le P_{i,t} \le U_{i,t}P_i^{1,l}, \text{ or, } \ldots, \text{ or} \\ &U_{i,t}P_i^{j-1,u} \le P_{i,t} \le U_{i,t}P_i^{j,l}, \text{ or, } \ldots, \text{ or} \\ &U_{i,t}P_i^{N_i,u} \le P_{i,t} \le U_{j,t}P_i^{\max} \quad \forall i \in PDG \end{aligned} \tag{1c}$$

where PDG represents a collection of the N PDGs in the considered SG, $f_{i,t}$ denotes PDG i's generation cost function at time t, $P_i^{\min}$ and $P_i^{\max}$ are the respective power output lower and upper bounds, $P_i^{j,l}$ and $P_i^{j,u}$ are the respective lower and upper bounds of the jth POZ of the ith PDG, N_i is the number of POZs of the ith PDG, $P_{d,t}$ is the total net load demand, and $P_{l,t}$ denotes the TLs. $U_{i,t}$ is a binary parameter. If the ith PDG is dispatchable at time t, then $U_{i,t} = 1$; otherwise, $U_{i,t} = 0$. Based on the Kron's method [12], the $P_{l,t}$ is formulated as follows:

$$P_{l,t} = \sum_{i=1}^{N} \sum_{j=1}^{N} P_{i,t} B_{ij} P_{j,t} + \sum_{i=1}^{N} B_{0i} P_{i,t} + B_{00} \tag{1d}$$

where B_{ij}, B_{0i}, and B_{00} are the coefficients.

Besides, it is assumed that the system involves VPEs [13] and MFOs [14]; then, not only the total net load demand $P_{d,t}$ but also the generation cost function $f_{i,t}(\cdot)$ are considered to be time varying as the price of fuel and generator changes. Therefore, the cost function in (1) is given as

$$f_{i,t}(P_{i,t}) = \begin{cases} \alpha_{1i,t}P_{i,t}^2 + \beta_{1i,t}P_{i,t} + \gamma_{1i,t} \\ \quad + \left|\varepsilon_{1i,t}\sin(\varphi_{1i,t}(P_i^{\min} - P_{i,t}))\right|, \; P_i^{\min} \le P_{i,t} \le P_{1i,t} \\ \alpha_{2i,t}P_{i,t}^2 + \beta_{2i,t}P_{i,t} + \gamma_{2i,t} \\ \quad + \left|\varepsilon_{2i,t}\sin(\varphi_{2i,t}(P_i^{\min} - P_{i,t}))\right|, \; P_{1i,t} \le P_{i,t} \le P_{2i,t} \\ \quad \ldots \\ \alpha_{mi,t}P_{i,t}^2 + \beta_{mi,t}P_{i,t} + \gamma_{mi,t} \\ + \left|\varepsilon_{mi,t}\sin(\varphi_{mi,t}(P_i^{\min} - P_{i,t}))\right|, \\ P_{(m-1)i,t} \le P_{i,t} \le P_i^{\max} \end{cases} \tag{1e}$$

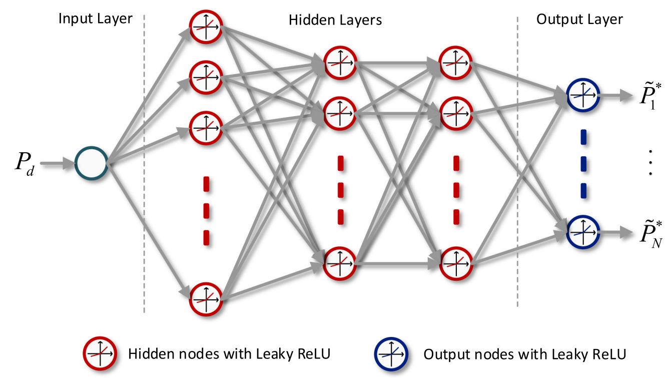

Fig. 1. DNN structure used in the preliminary approach. The fully connected neural network consists of one input layer, multiple hidden layers, and one output layer. Both the input layer and the output layer use leaky rectified linear unit (ReLU) as the activation function.

where $\alpha_{mi,t}$, $\beta_{mi,t}$, $\gamma_{mi,t}$, $\varepsilon_{mi,t}$, and $\varphi_{mi,t}$ are the ith PDG's cost coefficients for fuel type m at time t, and $p_{ni,t}$ is the corresponding fuel switching point.

Remark 1: We would like to emphasize that the optimization variables of the ED problem with *nonfixed* DGs are the outputs of all PDGs. If a PDG is not dispatchable at time t, i.e., $U_{i,t} = 0$, then its output is $P_{i,t} = 0$. Moreover, the power output of *inherently nondispatchable* generators (e.g., wind turbines and photovoltaic cells) has been integrated into $P_{d,t}$ [15].

Remark 2: Note that the ED problem with *nonfixed* DGs is essentially a discrete-time nonconvex optimization problem and, thus, can be solved by a variety of heuristic algorithms. However, it is well known that if the required time interval between two adjacent optimization operations is considerably small, then almost all heuristic algorithms are not competent for such a strict real-time requirement. Therefore, we urgently need a real-time ED algorithm to solve problem (1).

III. EDNet: A New Learning-Based ED Approach

In this section, we first introduce a preliminary DNN-based approach proposed in [16] to solve ED problems in *fixed* scenarios. To provide a distinguished ED solution for (1) in real time, we propose the EDNet by upgrading the DNN structure of the above approach in such a way that the developed EDNet can perfectly adapt to the *dynamic* SG system in terms of real time, simplicity, and generalization.

A. Preliminary DNN-Based Approach

Let us recall the ED problem with *fixed* DGs; given a total load demand P_d, it is required to find the corresponding optimal $P_i^*, \forall i$. Considering such an end-to-end mapping relationship, we can treat it as an "black box" and approximate it with a DNN. In fact, in [16], a fully connected DNN has been proposed to draw the above idea. The DNN used in the approach adopts a fully connected structure containing one input layer, multiple hidden layers, and one output layer, as illustrated in Fig. 1. It has been

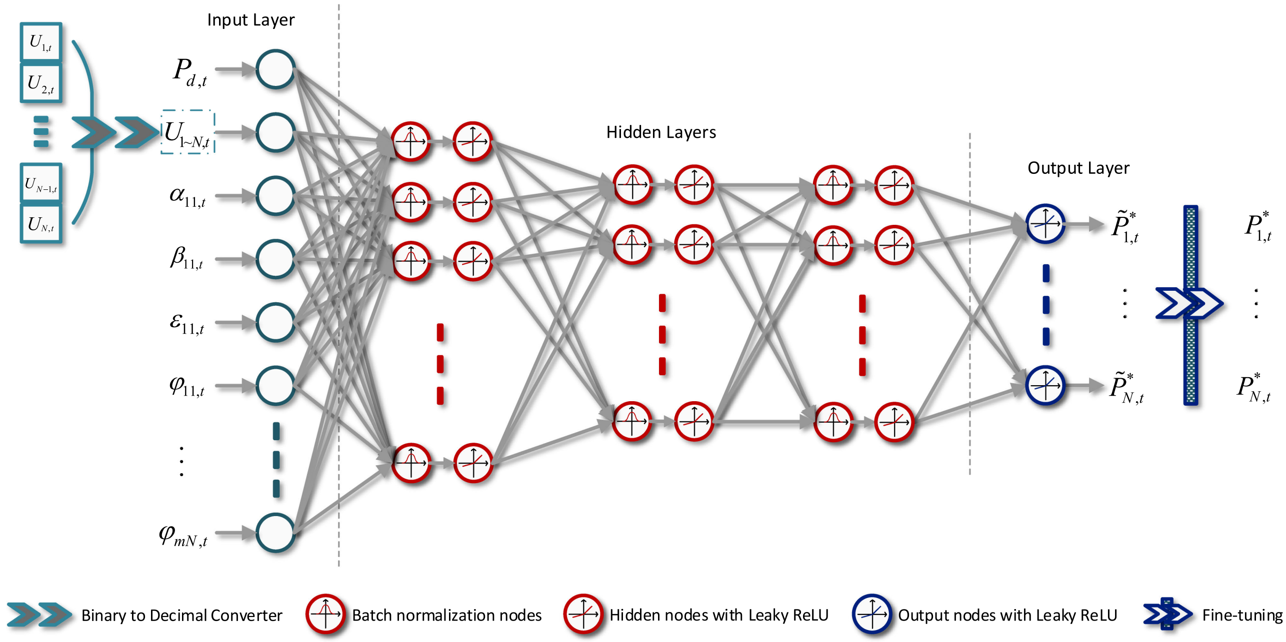

Fig. 2. DNN structure of the EDNet. The dispatchable states of N PDGs in an SG are converted into a 1-bit feature by a binary-to-decimal converter. The outputs of the EDNet are fine-tuned by a simple algorithm to incorporate the generation–demand balance constraint and local generation constraints.

shown that such a network possesses the capability of precisely approximating a hierarchical decentralized optimization algorithm [17] with a mean square error (MSE) between the network outputs and the ground truth less than 1e-3. Nevertheless, due to its limited input feature, such an approach seems to be helpless when facing a dynamic SG system including that we have considered in Section II-A. As a response, one intuitive solution is to separately train a range of DNNs for different dispatchable states and cost coefficients of PDGs. However, such a strategy is obviously susceptible to extremely high space complexity, i.e., the administrator needs to store a set of networks in the order of $\mathcal{O}(e^N)$ and select one in real time according to the practical scenario. To address this issue, a new real-time approach to efficiently solve (1) with satisfactory generalization capability is designed in this section.

B. Economic Dispatch Network

In this subsection, we propose the EDNet to tackle the above issue, as illustrated in Fig. 2. Our core idea is to cope with dynamic generation scenarios by training a single network instead of a collection of networks in such a way to meet the requirement of generalization while keeping the real-time capability and efficiency. Considering that the performance of a DNN inherently depends on the parameterizability to approximate the optimal ED function, we intend to introduce the EDNet in the aspects of network structure design, dataset generation, and training stage.

1) Network Design: Similar to the DNN structure used in the approach introduced in Section III-A, a fully connected DNN structure is exploited for the EDNet, as illustrated in Fig. 2. The network is comprised of one input layer containing $(mN+2)$ nodes, multiple hidden layers, and one output layer containing N nodes. To capture the dynamics of the PDGs in an SG system, the dispatchable states and the cost coefficients of all PDGs, in addition to the total net load demand $P_{d,t}$, are taken as the inputs of the network. Especially, the dispatchable state, which is an N-bit binary vector, has been converted into a 1-bit decimal scalar and being a part of features to be fed. The rationale behind such a preprocessing operation is to save the calculation time by reducing the number of nodes in the input layer [18]. It is worth noting that the cost coefficients $\gamma_{mi,t}, \forall m, i$ are not included in the input features owing to the fact that they do not make any impact on the ED optimization results.

In addition, considering that the mapping relationship to be learned by the EDNet is much more complicated than that shown in Fig. 1, we add a batch normalization (BN) layer before each hidden layer to accelerate the training process [19]. Moreover, the output vector of the EDNet $[\tilde{P}^*_{1,t},\ldots,\tilde{P}^*_{N,t}]^\top$ may be in slightly conflict with local generation constraints and supply–demand balance constraint so that it requires some fine adjustment. In this article, a simple algorithm [21] is employed to make the solution obtained by the EDNet to be feasible, as illustrated in the following.

First, we detect the average generation–demand mismatch of the whole SG system by

$$\delta[n] = \frac{P_{d,t} + P_{l,t} - \sum_{i \in DG} \tilde{P}^*_{i,t}[n]}{N} \tag{2}$$

where n denotes the iteration index and $\tilde{P}^*_{i,t}[0] = \tilde{P}^*_{i,t}$. Then, we update the output variables

$$\tilde{P}^*_{i,t}[n+1] = \tilde{P}^*_{i,t}[n] + \delta[n] \quad \forall i \tag{3}$$

and conduct projection operation to meet the local constraint, $\forall i$

$$\tilde{P}^*_{i,t}[n+1] = \begin{cases} P_i^{j,l}, & \text{if } P_i^{j,l} < \tilde{P}^*_{i,t}[n+1] \leq \frac{P_i^{j,l}+P_i^{j,u}}{2} \\ P_i^{j,u}, & \text{if } \frac{P_i^{j,l}+P_i^{j,u}}{2} < \tilde{P}^*_{i,t}[n+1] < P_i^{j,u} \\ \tilde{P}^*_{i,t}[n+1], & \text{otherwise} \end{cases}. \tag{4}$$

Let $n = n + 1$ and continue the iteration steps in (2)–(4) until $\delta[n] < \sigma$, which indicates that the balance between supply and demand is achieved. Note that σ is a tolerance, which is usually set to be very small. If the stopping iteration index is denoted as n_s, then we obtain the optimal solution

$$[P^*_{1,t}, \ldots, P^*_{N,t}]^\top = [\tilde{P}^*_{1,t}[n_s], \ldots, \tilde{P}^*_{N,t}[n_s]]^\top. \tag{5}$$

Since a single forward propagation of the EDNet mainly requires several layers of matrix–vector multiplications, its time complexity can be expressed as $\mathcal{O}(mNn_l n_n^2)$, where n_l and n_n denote the number of layers and the number of neurons in a single layer, respectively. Besides, inspired by the approach in [23], the time complexity of the fine-tuning algorithm (2)–(5) can be described as $\mathcal{O}(\log_2 \frac{1}{\sigma})$. Therefore, the total runtime required to obtain the optimal solution through the EDNet can be denoted as $\mathcal{O}(mNn_l n_n^2 + \log_2 \frac{1}{\sigma})$.

Remark 3: We should mention that there are several alternative DNN architectures, such as convolutional neural network, recurrent neural network, and graph neural network. However, we easily find that there lacks a special feature structure in problem (1) that can be exploited for such advanced feature extraction means, including *convolution* and *graph embedding*, which motivates us to choose a concise and general fully connected structure in this article.

Remark 4: It is worth pointing out that the proposed approach is also applicable if the *ramp rate* constraints are considered in problem (1), which results in a dynamic ED problem. A feasible solution is to incorporate the power output of all PDGs at the previous time slot into the input features. The feasibility is guaranteed by the determined mapping relationship between the input features and the optimal solution, while EDNet can be employed to approximate the mapping.

Remark 5: As the mapping relationship to be expressed by the EDNet is so complicated, it is necessary to introduce the BN layer into our network. A lot of literature has studied how BN accelerates the training of the DNN and avoids vanishing the gradient problem [19]. What we want to emphasize here is that as an approach that may be applied to the practical power grid, reducing training time will bring substantial convenience to the operators. Besides, the ensemble framework to be proposed in the next section, which includes multiple EDNets, will also greatly benefit from the role of the BN layer.

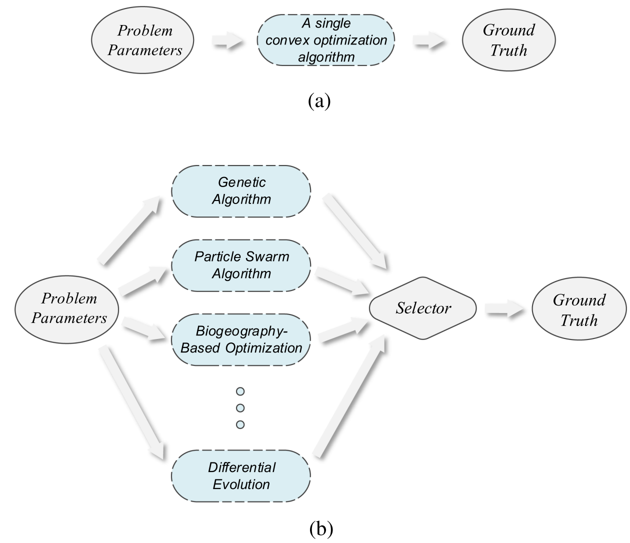

Fig. 3. Comparison illustration between the methods for searching the ground truth of convex and nonconvex ED problems. (a) Typical approach for obtaining the ground truth of convex ED problems. (b) Proposed ensemble framework for exploring the ground truth of nonconvex ED problems.

2) Dataset Generation: Typically, the dataset should contain a sufficient representation of the ground truth so that the network can be trained to approximate the best performance. As discussed earlier, if only a single nonconvex ED algorithm is utilized to explore the ground truth, the trained network may not always provide better performance than other nonconvex ED algorithms. Therefore, in this work, we advocate using the dataset obtained by *ensembling* several state-of-the-art effective algorithms in solving nonconvex ED problems, as shown in Fig. 3(b). Specifically, first, the problem parameters (e.g., total load demand, unit status, and cost coefficients) of a given nonconvex ED problem are fed into these algorithms in parallel. Second, each deployed algorithm solves this nonconvex ED problem separately and provides its solution to the selector. Note that for some deployed heuristic algorithms containing a series of random strategies, the solution should be the optimal result of multiple trials. Finally, after collecting the solutions from all deployed algorithms, the solution with the least power generation cost is selected as the ground truth. Repeatedly executing this ensemble framework with arbitrary distinct problem parameters, we can finally get the entire dataset $(P_{d,t}{}^{(k)}, U_{1-N,t}{}^{(k)}, \ldots, \varphi_{mN,t}{}^{(k)}, \{P^*_{i,t}{}^{(k)}\}_{i=1}^N)_{k=1}^K$, where k indexes the data sample. After that, the dataset is randomly divided into a training set, a validation set, and a test set at a ratio of 7:1:2 with reference to commonly used training strategies [18]. Moreover, all the training data have been normalized into $[-1, 1]$ to accelerate the training process, which is also simultaneously operated on the validation and test data. Note that due to our aim of approximating the mapping of problem (1) with only one network, the high space complexity in the order of $\mathcal{O}(e^N)$ has been transferred from the number of networks (as discussed in Section III-A) to the scale of the training data for such a single network. Also, it is important to note that a sufficiently representative training dataset is necessary in our work. In other words, the mapping space of problem (1) should be covered by a finite number of training samples as much as possible so that the well-trained EDNet will show an excellent generalization capability to the whole mapping space theoretically [18].

3) EDNet Training: Based on the prepared training set, we use the backpropagation technique to continuously optimize the weights of the EDNet so that its outputs gradually approach the ground truth $\{P_{i,t}^{*}{}^{(k)}\}_{i=1}^{N}$. The employed loss function is the MSE between the ground truth and the outputs of the EDNet, i.e.,

$$L(\Theta) = \frac{\sum\limits_{k}\sum\limits_{i=1}^{N} \|\tilde{P}_{i,t}^{*}{}^{(k)}(I^{(k)},\Theta) - P_{i,t}^{*}{}^{(k)}\|^2}{K_{\text{tr}}} \tag{6}$$

where K_{tr} is the number of training samples. $\tilde{P}_{i,t}^{*}{}^{(k)}(I^{(k)},\Theta)$ is the approximate optimal ED solution obtained by the EDNet with the optimized weights Θ and the input features $I^{(k)} = \{P_{d,t}{}^{(k)},\ldots,\varphi_{mN,t}{}^{(k)}\}$. The optimizer we employ for such an approximation task is *adam*, which has been widely acclaimed by researchers recently due to its outstanding efficiency and adaptability for a variety of complex optimization problems [25]. Besides, to further improve the training performance, the batch size and the learning rate are selected by cross validation, and the *Xavier* normal initializer is adopted for the initialization of network weights Θ [26]. The training process stops as long as the validation set's MSE value is lower than a set threshold value s.

Remark 6: Note that in the training process of the EDNet, we focus on minimizing the *average* loss (6) between the ground truth and the outputs of the EDNet, which indicates that there are some "hard samples" resulting in that a small amount of mapping space of problem (1) is approximated with relatively large errors (e.g., dozens of times larger than the mean approximation errors) by a single EDNet. Obviously, when some particular test samples are fed into the EDNet, it may yield nonideal solutions with too large generation cost. Such a limitation gives rise to the ensemble EDNet.

IV. E^2 DNet: Ensembling EDNets

Motivated by the limitation discussed in Remark 6, we integrate our EDNet with the idea of *ensemble learning* and propose a novel ED scheme, called E^2 DNet, as illustrated in Fig. 4. In such an ensemble framework, each EDNet plays a role of *weak* learner, and their opinions are collected to form a *strong* ED solution. As pointed out in [27], the *diversity* of these weak learners is an essential characteristic of ensemble learning. Hence, in this article, an *AdaBoost* (adaptive boosting) technique [28] is utilized to train several EDNets arranged in parallel and guarantee the uniqueness of each EDNet. Considering that the performance of E^2 DNet largely depends on the training details, we will describe the proposed E^2 DNet in terms of its training stage and testing stage.

Algorithm 1: Simplified Adaboost.R2 Algorithm for Training E^2 DNet.

Input:
- Given K_{tr} training samples $(I^{(k)},\{P_{i,t}^{*}{}^{(k)}\}_{i=1}^{N})_{k=1}^{K_{\text{tr}}}$.
- Number of iterations (weak learners) J.
- Weak learner $\Theta_j,\ j=1,\ldots,J$.

Initialize:
- Distribution $D_j(k) = 1/K_{\text{tr}}$ for all k.
- Iteration $j=1$.
- Error rate $\overline{\epsilon_j} = 0$.

for $j = 1$ to J **do**

S1: Train Θ_j with the training samples under distribution $D_j(k)$.

S2: Calculate the absolute error for each training sample as:
$\epsilon_{\text{abs},j}(k) = \sum_{i=1}^{N} \|\tilde{P}_{i,t}^{*}{}^{(k)}(I^{(k)},\Theta_j) - P_{i,t}^{*}{}^{(k)}\|^2$.

S3: Calculate the relative error for each training sample as: $\epsilon_{\text{rel},j}(k) = \frac{\epsilon_{\text{abs},j}(k)}{\max_{k=1,\ldots,K_{\text{tr}}}\epsilon_{\text{abs},j}(k)}$.

S4: Calculate the error rate for the jth weak learner:
$\overline{\epsilon_j} = \sum_{k=1}^{K_{\text{tr}}} D_j(k)\epsilon_{\text{rel},j}(k)$.

if $\overline{\epsilon_j} > 1/2$ **then**
 break
end if

S5: Update distribution for the next weak learner:
$D_{j+1}(k) = \frac{D_j(k)(\frac{\overline{\epsilon_j}}{1-\overline{\epsilon_j}})^{1-\epsilon_{\text{rel},j}(k)}}{Z_j}$, where Z_j is a normalization factor such that D_{j+1} will be a distribution.

end for

Output: $\arg\min_{\Theta_j} \sum_{i=1}^{N} f_{i,t}(P_{i,t}^{*}(I,\Theta_j))$, where $P_{i,t}^{*}(I,\Theta_j)$ is the fine-tuned solution (refer to Formula (2)–(5) or Fig. 2) obtained by Θ_j with a particular input I.

A. Training Stage

In fact, the Boosting family comprises a series of members that could be useful for our approximation task. Based on the classic AdaBoost.R2 [28], we propose a simplified variant named Simplified Adaboost.R2 as the training strategy in our work, as shown in Algorithm 1. The rationale behind the algorithm is to train a series of weak learners in order while letting the "hard samples" recognized by the current weak learner gain a higher proportion in training the subsequent weak learners. In other words, our proposed algorithm is to construct a strong learner, where each weak learner shows a powerful approximation to a part of the mapping space of problem (1) so that their combination achieves a globally excellent approximation to the entire mapping space. Recalling that our essential goal is to pick the one with the least generation cost among the several solutions obtained by all the weak learners, we remove the measure of

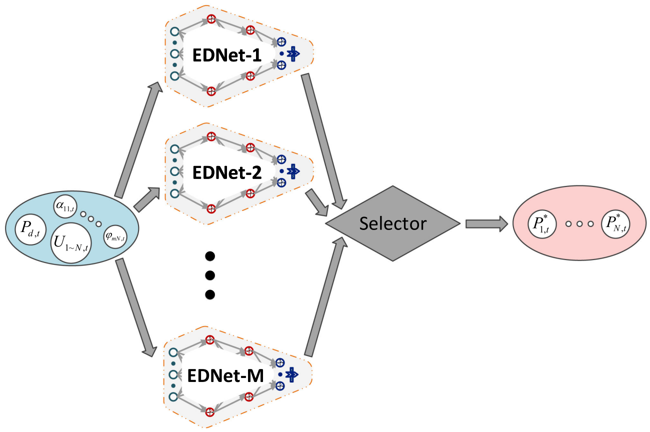

Fig. 4. Scheme of the E^2 DNet.

confidence in the weak learner (or weight of the weak learner) contained in the original AdaBoost.R2 algorithm because it is redundant to our problem. In addition, the output hypothesis of Simplified AdaBoost.R2 should be the weak learner generating solution with the least generation cost rather than the one selected according to the weighted median rule in the original AdaBoost.R2 [28]. Note that the time complexity of Algorithm 1 can be expressed as $\mathcal{O}(mNn_l n_n^2 K_{\text{tr}} TJ)$, where T denotes the number of iterations required by the optimizer for training a single EDNet.

B. Testing Stage

In the testing stage, the features are separately fed into each EDNet, while the selector collects the solutions from all EDNets and select the best one with the least generation cost. Note that the proposed EDNet just needs several operations such as matrix multiplications, which indicates a low computation complexity. Even though the solutions are calculated consecutively in the E^2 DNet due to the limitation of hardware or programming, the computation complexity of E^2 DNet is only linearly increased with the number of the EDNets in the ensemble. Furthermore, we advocate using parallel computing in the testing stage (or practical running stage), so that the computational complexity of the E^2 DNet is almost the same as that of a single EDNet, which can be expressed as $\mathcal{O}(mNn_l n_n^2 + \log_2 \frac{1}{\sigma})$.

Remark 7: We would like to point out that our work is similar in form but quite different in substance from the ensemble framework developed in [30]. Such a framework employs the optimizer such as *adam* to directly solve an interference management optimization problem in wireless communication, which is essentially an *unsupervised*-learning-based approach, whereas our work is a typical *supervised* learning with the solutions obtained by the state-of-the-art algorithms as the ground truth. Moreover, the involved ensemble framework in [30] is designed to collect a variety of local optimal solutions from the weak learners, while our work aims to let all weak learners place emphasis on approximating distinct (may partly superposed) part of the mapping space of problem (1).

Remark 8: We should emphasize that an E^2 DNet is trained to solve ED problems of one specific power system. In other words, for another power system with different characteristics (e.g., system scale, generation cost function, and constraints), the ground truth should be re-explored and collected, and another E^2 DNet needs to be trained.

Remark 9: Different from conventional learning-based ED (or some other tasks) approaches, the presented ensemble framework simultaneously provides advantages in terms of real time, conciseness, generalization, and, most importantly, better solution performance, which is guaranteed by the inherent properties of ensemble learning. Such advantages will be verified in the next section.

V. Case Studies

A. Test System Selection

In this section, to verify the performance of our proposed EDNet and E^2 DNet, a ten-unit power system with MFOs and VPEs is adopted as our test system, whose characteristics can be found in [20]. Note that two fuel types are provided for the first generation unit, and three types for the others.

B. Baselines

The algorithms in [20], [22], [24], [29], and [31], which are widely used in many research works, have been used in this system as a benchmark test. All the related code (including implemented baselines and E^2 DNet) can be found at.[2]

C. Network Setup

1) Dataset Generation: Based on the above ten-unit test system, we use an ensemble of GA, PSO, BBO, DE, and L-MILP to generate a dataset containing 1 000 000 samples, which is further randomly divided into a training set (700 000 samples), a validation set (200 000 samples), and a test set (100 000 samples). In the input (feature) part of each sample, $U_{1\sim N,t}$ traverses all possible unit status of ten units, $P_{d,t}$ is randomly sampled from the corresponding feasible region, and the power generation coefficients of the units are randomly sampled from [0.9,1.1] times the standard coefficients provided in [20]. In addition, the output (solution) part of each sample is the optimal solution obtained by the ensemble algorithms.

2) Network Training: Through trial and error, the values of tuning parameters in this work are set as follows: The EDNets contain one input layer, ten hidden layers (including BN layers), and one output layer, where each hidden layer contains 200 neurons. The batch size and the learning rate are selected as 1000 and 1e-3, respectively. The stopping criterion is set to $s = 1e - 3$. The ensembling size of E^2 DNet is set to $M = 10$. The other preliminary training details have been mentioned in Section III-B3. All the training tasks in this article are carried out with TensorFlow 1.13.1 supported by two ten-core Intel i7-6950X 3.0-GHz CPUs, two Nvidia K40 GPUs, and 128 GB of memory.

[2][Online]. Available: https://github.com/599143868/E-2DNet-TII

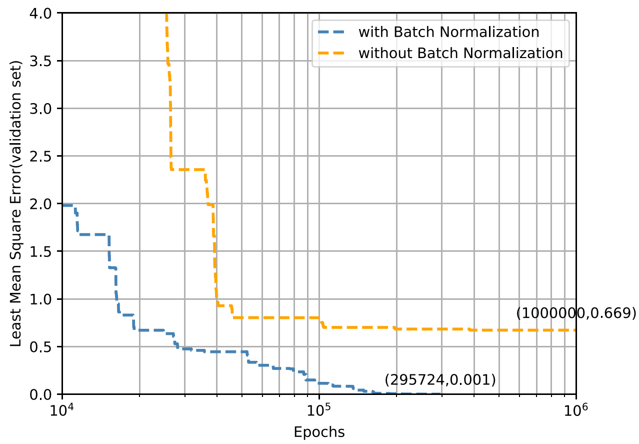

Fig. 5. Convergence performance of the EDNet trained with and without BN.

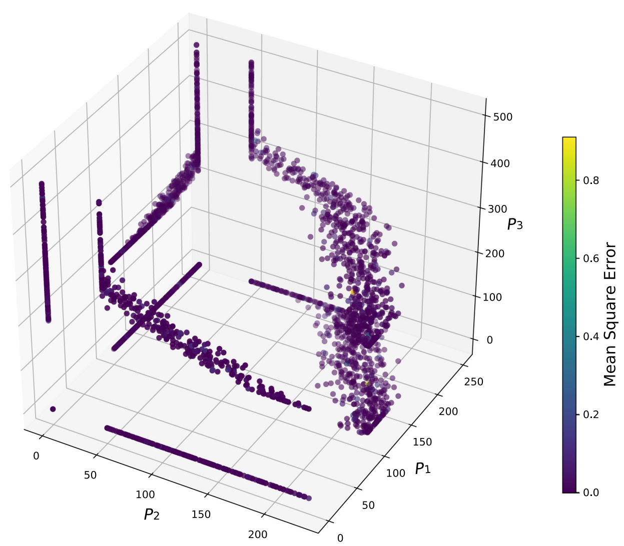

Fig. 6. Approximation performance of the EDNet.

As discussed in Section III-B3, BN makes a significant contribution to improving the training efficiency. Therefore, we plot the least MSE curve of the validation set to compare the training efficiency of the EDNet with and without BN, as shown in Fig. 5. It can be observed that the training of the EDNet with BN is completed at the 295 724th epoch, while the one without BN distinctly suffers from a slow convergence rate as the least MSE of the validation set is still 0.669 at the 1e6th epoch.

D. Network Approximation Performance

To validate the approximation performance of a single EDNet, 3000 test samples are randomly selected from the test set and fed into the well-trained EDNet to obtain the solutions for visualization. As shown in Fig. 6, each point in the 3-D space represents a solution obtained by L-MILP, while its color indicates the MSE between it and the solution obtained by the EDNet. It can be observed that the optimal solutions found by our proposed EDNet are very close to those obtained by L-MILP,

TABLE I
PARAMETER SETTINGS OF EIGHT DESIGNED CASES

Case	Load Demand (MW)	Unit Status	Cost Coefficients	Constraints
1	2700	All on	Nominal value	VPE and MFOs
2	3000	All on	Nominal value	
3	2700	DG 2 is off	Nominal value	
4	2700	All on	10% perturbation on all cost coefficients	
5	2700	All on	Nominal value	VPE, MFOs, POZs and TLs
6	2750	All on	Nominal value	
7	2700	DG 5 is off	Nominal value	
8	2700	All on	5% perturbation on all cost coefficients	

with the average MSEs being 1.02e-3 over the 3000 test samples. However, still several solutions are plotted in yellow or blue, indicating that there exists a relatively larger approximation error in a smaller proportion of the solutions produced by the EDNet, which corroborates Remark 6 in Section III-B3.

E. Algorithm Performance

1) Performance on Minimizing the Overall Generation Cost: To verify such a performance and scalability of our proposed EDNet and E^2 DNet in different scenarios, we design four cases (i.e., Cases 1–4) with different power load demand $P_{d,t}$, unit status $U_{1\sim N,t}$, and cost coefficients, as shown in Table I. The detailed power allocation results given by the proposed E^2 DNet are shown in Table II, while the comparison results in terms of generation cost are shown in Table III. It is observed that the proposed E^2 DNet surpasses most baselines.

With POZs and TLs further considered in this power system, four cases (i.e., Cases 5–8) are designed, and the detailed settings are illustrated in Table I. The involved coefficients of POZs and TLs are taken from [22]. The data generation and network training are similar to those mentioned in Section V-C. The detailed power allocation results provided by the proposed E^2 DNet are shown in Table IV, and the comparison results in terms of generation cost are shown in Table V. As an overview, we can readily find that the proposed E^2 DNet outperforms most baseline algorithms. Specifically, first of all, it can be observed that the proposed EDNet and E^2 DNet achieve less generation cost than the DNN [8] in Cases 1, 2, 5, and 6. There are two main reasons behind this: 1) the developed ensemble exploration framework provides better ground truth than the traditional approach in [8], revealing its superiority in exploring ground truth for nonconvex problems and 2) the designed network structure enables EDNet and E^2 DNet to express complex nonlinear mapping relationships more accurately than the DNN [8]. Second, we find that the EDNet and the E^2 DNet show satisfactory performance in Cases 3, 4, 7, and 8, whereas the DNN [8] is inapplicable in these cases since its network design determines that it cannot handle dynamic unit status and cost coefficients. Besides, it can be seen that the E^2 DNet slightly outperforms a single EDNet, revealing the enhancement effect brought by our ensemble framework.

2) Real-Time Performance: To verify the real-time performance of the proposed E^2 DNet, three power systems consisting of 10, 40, and 120 units are considered. Note that the 40- and

TABLE II
SOLUTION FOUND BY THE E^2 DNET IN FOUR CASES OF THE TEN-UNIT SYSTEM WITH VPE AND MFOS

DG	Case 1 Power Allocation	Case 1 Fuel type	Case 2 Power Allocation	Case 2 Fuel type	Case 3 Power Allocation	Case 3 Fuel type	Case 4 Power Allocation	Case 4 Fuel type
1	218.1147	2	227.3235	2	238.6108	2	218.5473	2
2	211.1581	1	217.8485	1	0	-	211.5956	1
3	280.6484	1	499.7031	2	308.8923	1	280.6571	1
4	239.6887	3	242.5082	3	246.4076	3	239.6394	3
5	279.8146	1	294.3240	1	315.1694	1	278.2697	1
6	239.6618	3	242.4814	3	247.7242	3	239.6516	3
7	287.7275	1	304.3246	1	346.8316	2	288.3741	1
8	239.6842	3	243.1801	3	245.7320	3	239.6028	3
9	427.6333	3	439.9512	3	439.9954	1	428.9432	3
10	275.8686	1	288.3553	1	310.6366	1	274.7191	1
Total generation cost ($/h)	623.8583		785.9001		701.9744		686.2437	

TABLE III
COMPARISON OF PERFORMANCE ON MINIMIZING THE OVERALL GENERATION COST BETWEEN E^2 DNET AND OTHER BASELINES ON THE TEN-UNIT SYSTEM WITH VPE AND MFOS

		Total generation cost ($/h)											
Algorithm	Trials	Case 1			Case 2			Case 3			Case 4		
		Min	Avg	Max	Min	Avg	Max	Min	Avg	Max	Min	Avg	Max
GA [20]	100	625.37	627.81	630.95	787.79	795.18	803.70	702.82	706.80	713.56	688.28	694.24	709.53
PSO [22]	100	623.95	624.51	638.35	785.99	788.67	797.24	702.02	702.73	705.05	686.29	688.42	701.21
BBO [24]	100	623.90	624.10	626.46	785.92	786.13	792.46	701.98	702.09	702.38	686.31	686.44	689.06
DE [29]	100	623.91	624.07	626.50	785.96	786.08	786.98	702.01	702.10	702.32	686.32	686.43	687.08
L-MILP [31]	1		623.86			785.90			702.05			686.24	
DNN [8]	1		623.95			785.99			inapplicable			inapplicable	
EDNet	1		**623.86**			**785.91**			**702.00**			**686.24**	
E^2DNet	1		**623.86**			**785.90**			**701.98**			**686.24**	

TABLE IV
SOLUTION FOUND BY THE E^2 DNET IN FOUR CASES OF THE TEN-UNIT SYSTEM WITH VPE, MFOS, POZS, AND TLS

DG	Case 5 Power Allocation	Case 5 Fuel type	Case 6 Power Allocation	Case 6 Fuel type	Case 7 Power Allocation	Case 7 Fuel type	Case 8 Power Allocation	Case 8 Fuel type
1	224.2767	2	229.4117	2	235.5634	2	224.0439	2
2	214.3843	1	216.6056	1	217.8458	1	215.7515	1
3	286.7063	1	293.7638	1	499.6836	2	286.4264	1
4	240.8980	3	241.9730	3	244.2579	3	240.7514	3
5	287.1293	1	297.8630	1	0	-	286.6875	1
6	250.0084	3	250.0084	3	264.6046	3	250.0838	3
7	290.0000	1	290.0000	1	331.1132	2	290.0000	1
8	250.0308	3	250.0308	3	264.4378	3	250.1359	3
9	421.4526	3	437.4310	3	276.1971	1	423.6688	3
10	269.5462	1	279.0298	1	406.8051	3	266.8322	1
Transmission losses (MW)	34.4327		36.1171		40.5085		34.3815	
Total generation cost ($/h)	642.8653		669.8293		760.7133		675.0705	

TABLE V
COMPARISON OF PERFORMANCE ON MINIMIZING THE OVERALL GENERATION COST BETWEEN E^2 DNET AND OTHER BASELINES ON THE TEN-UNIT SYSTEM WITH VPE, MFOS, POZS, AND TLS

		Total generation cost ($/h)											
Algorithm	Trials	Case 5			Case 6			Case 7			Case 8		
		Min	Avg	Max	Min	Avg	Max	Min	Avg	Max	Min	Avg	Max
GA [20]	100	644.88	651.70	668.17	671.61	678.23	690.73	762.68	767.14	775.28	676.40	684.75	696.74
PSO [22]	100	642.93	643.30	654.19	669.88	671.16	678.12	760.77	762.38	765.03	675.07	675.71	686.39
BBO [24]	100	642.93	643.39	644.85	669.94	670.12	673.12	760.70	760.76	760.85	675.10	675.57	677.10
DE [29]	100	642.97	643.08	643.88	670.04	670.14	670.72	760.73	760.90	762.23	675.15	675.25	675.74
L-MILP [31]	1		642.87			669.83			760.77			675.18	
DNN [8]	1		643.02			670.08			inapplicable			inapplicable	
EDNet	1		**642.90**			**669.83**			**760.73**			**675.08**	
E^2DNet	1		**642.87**			**669.83**			**760.71**			**675.07**	

120-unit systems are obtained by duplicating four times and 12 times the ten-unit system, respectively. Table VI reports the CPU time required by our proposed E^2 DNet and other baselines on the three systems. First, we can observe that the proposed E^2 DNet requires much less CPU time than other baselines on these three systems. Second, it is only milliseconds that the proposed E^2 DNet need to give a similar ED result. In contrast, the CPU time required by other baselines significantly increases as the system scales up. For instance, the L-MILP even takes more than 10 000 s to complete such an optimization task on the 120-unit system. We easily find that the proposed E^2 DNet achieves orders of magnitude computational speedup compared

TABLE VI
COMPARISON OF COMPUTATIONAL PERFORMANCE BETWEEN E^2 DNET AND OTHER BASELINES

	Algorithm	GA (100 trials) (MATLAB)	PSO (100 trials) (MATLAB)	BBO (100 trials) (MATLAB)	DE (100 trials) (MATLAB)	L-MILP (MATLAB)	**EDNet** (Python)	**E^{2}DNet** (Python)
10-unit	Total CPU time (sec.)	8.50	4.88	34.77	6.12	23.90	**1.28e-3**	**2.91e-3**
	Times	2.92e3×	1.68e3×	1.19e4×	2.10e3×	8.21e3×	**0.44**×	**1**×
40-unit	Total CPU time (sec.)	21.75	16.50	185.98	18.69	409.91	**1.48e-3**	**3.62e-3**
	Times	6.01e3×	4.56e3×	5.14e4×	5.16e3×	1.13e5×	**0.41**×	**1**×
120-unit	Total CPU time (sec.)	2041.15	1652.58	1526.47	2143.23	25982.14	**1.87e-3**	**3.89e-3**
	Times	5.25e5×	4.25e5×	3.92e5×	5.51e5×	6.68e6×	**0.48**×	**1**×

to other baselines. It is not a surprising result since only several layers of simple operations such as matrix multiplications are involved in the computation of the proposed approach.

VI. CONCLUSION

In this article, we presented a novel ensemble learning-based approach to solve the nonconvex ED problem in real time. First, the *EDNet* was proposed to learn the complicated mapping space of a nonconvex ED problem considering varying total load demand, generator status, and generation cost functions. Since a single EDNet may not show perfect response to every part of the mapping space, we further presented *E^2 DNet*, an ensemble of EDNets, which is prone to improve the performance on minimizing the overall generation cost while keeping all the excellence of EDNet in simplicity, real time, and generalization. A significant advantage of the proposed approach is that it can achieve orders of magnitude computational speedup while guaranteeing the performance on minimizing the overall generation cost compared to the state-of-the-art nonconvex ED algorithms. Besides, compared with some existing DNN-based ED approaches, the proposed approach can tackle dynamic operating scenarios including varying power load demand, unit status, and generation cost functions. We believe that the power of such an ensemble framework will also be revealed in some other optimization problems far beyond ED.

There are some challenging but interesting works remaining to be studied in future. First, the proposed method requires a lot of training data and considerable training time (in hours) in the offline training stage. Therefore, how to reduce the training cost is an important future research direction. Second, how to build distributed variants of EDNet and E^2 DNet so that they can be applied in the distributed framework of SG is still an open problem. Last but not the least, how to use our "learning to optimize" idea to solve more complex SG optimization tasks, such as unit commitment, energy storage optimization, and robust optimization, will also be considered in our future studies.

REFERENCES

[1] A. Askarzadeh, "A memory-based genetic algorithm for optimization of power generation in a microgrid," *IEEE Trans. Sustain. Energy*, vol. 9, no. 3, pp. 1081–1089, Jul. 2018.

[2] J. Zhao, S. Liu, M. Zhou, X. Guo, and L. Qi, "Modified cuckoo search algorithm to solve economic power dispatch optimization problems," *IEEE/CAA J. Autom. Sinica*, vol. 5, no. 4, pp. 794–806, Jul. 2018.

[3] Q. Zhu, Z. Chen, and Y. C. Soh, "A novel semisupervised deep learning method for human activity recognition," *IEEE Trans. Ind. Informat.*, vol. 15, no. 7, pp. 3821–3830, Jul. 2019.

[4] Q. Li, L. Li, Q. Li, and J. Zhong, "A comprehensive exploration on spider with fuzzy decision text-to-SQL model," *IEEE Trans. Ind. Informat.*, vol. 16, no. 4, pp. 2542–2550, Apr. 2020.

[5] W. Fang, F. Xue, Y. Ding, N. Xiong, and V. Leung, "EdgeKE: An on-demand deep learning IoT system for cognitive big data on industrial edge devices," *IEEE Trans. Ind. Informat.*, vol. 17, no. 9, pp. 6144–6152, Sep. 2021.

[6] K. Hornik, M. Stinchcombe, and H. White, "Multilayer feedforward networks are universal approximators," *Neural Netw.*, vol. 2, no. 5, pp. 359–366, 1989.

[7] J. Nanda *et al.*, "Application of artificial neural network to economic load dispatch," in *Proc. 4th Int. Conf. Adv. Power Syst. Control, Oper. Manage.*, 1997, pp. 707–711.

[8] Y. Yang, Z. Yang, J. Yu, K. Xie, and L. Jin, "Fast economic dispatch in smart grids using deep learning: An active constraint screening approach," *IEEE Internet Things J.*, vol. 7, no. 11, pp. 11030–11040, Nov. 2020.

[9] F. Guo, B. Xu, W.-A. Zhang, C. Wen, D. Zhang, and L. Yu, "Training deep neural network for optimal power allocation in islanded microgrid systems: A distributed learning-based approach," *IEEE Trans. Neural Netw. Learn. Syst.*, to be published, doi: 10.1109/TNNLS.2021.3054778.

[10] F. Guo, B. Xu, L. Xing, W.-A. Zhang, C. Wen, and L. Yu, "An alternative learning-based approach for economic dispatch in smart grid," *IEEE Internet Things J.*, to be published, doi: 10.1109/JIOT.2021.3072840.

[11] F. Li, J. Qin, and W. X. Zheng, "Distributed Q-learning-based online optimization algorithm for unit commitment and dispatch in smart grid," *IEEE Trans. Cybern.*, vol. 50, no. 9, pp. 4146–4156, Sep. 2020.

[12] A. J. Wood, B. F. Wollenberg, and G. B. Sheblé, *Power Generation, Operation, and Control*. Hoboken, NJ, USA: Wiley, 2013.

[13] G. Binetti, A. Davoudi, D. Naso, B. Turchiano, and F. L. Lewis, "A distributed auction-based algorithm for the nonconvex economic dispatch problem," *IEEE Trans. Ind. Informat.*, vol. 10, no. 2, pp. 1124–1132, May 2014.

[14] A. Meng, J. Li, and H. Yin, "An efficient crisscross optimization solution to large-scale non-convex economic load dispatch with multiple fuel types and valve-point effects," *Energy*, vol. 113, pp. 1147–1161, 2016.

[15] S.-J. Ahn, S.-R. Nam, J.-H. Choi, and S.-I. Moon, "Power scheduling of distributed generators for economic and stable operation of a microgrid," *IEEE Trans. Smart Grid*, vol. 4, no. 1, pp. 398–405, Mar. 2013.

[16] F. Guo, B. Xu, W.-A. Zhang, D. Zhang, and L. Yu, "Real-time optimal power allocation for smart grid system via deep neural network: A learning based approach," in *Proc. Chin. Control Decis. Conf.*, 2020, pp. 868–873.

[17] F. Guo, C. Wen, J. Mao, J. Chen, and Y.-D. Song, "Hierarchical decentralized optimization architecture for economic dispatch: A new approach for large-scale power system," *IEEE Trans. Ind. Informat.*, vol. 14, no. 2, pp. 523–534, Feb. 2018.

[18] Y. LeCun, Y. Bengio, and G. Hinton, "Deep learning," *Nature*, vol. 521, no. 7553, pp. 436–444, 2015.

[19] S. Ioffe and C. Szegedy, "Batch normalization: Accelerating deep network training by reducing internal covariate shift," 2015, *arXiv:1502.03167*.

[20] C.-L. Chiang, "Improved genetic algorithm for power economic dispatch of units with valve-point effects and multiple fuels," *IEEE Trans. Power Syst.*, vol. 20, no. 4, pp. 1690–1699, Nov. 2005.

[21] F. Li, J. Qin, Y. Kang, and W. X. Zheng, "Consensus based distributed reinforcement learning for nonconvex economic power dispatch in microgrids," in *Proc. Int. Conf. Neural Inf. Process.*, 2017, pp. 831–839.

[22] Z.-L. Gaing, "Particle swarm optimization to solving the economic dispatch considering the generator constraints," *IEEE Trans. Power Syst.*, vol. 18, no. 3, pp. 1187–1195, Aug. 2003.

[23] J. Deng and Q. Zhang, "Combining simple and adaptive Monte Carlo methods for approximating hypervolume," *IEEE Trans. Evol. Comput.*, vol. 24, no. 5, pp. 896–907, Oct. 2020.

[24] A. Bhattacharya and P. K. Chattopadhyay, "Hybrid differential evolution with biogeography-based optimization for solution of economic load dispatch," *IEEE Trans. Power Syst.*, vol. 25, no. 4, pp. 1955–1964, Nov. 2010.

[25] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," 2014, *arXiv:1412.6980*.

[26] X. Glorot and Y. Bengio, "Understanding the difficulty of training deep feedforward neural networks," in *Proc. 13th Int. Conf. Artif. Intell. Statist.*, 2010, pp. 249–256.

[27] C. Zhang and Y. Ma, *Ensemble Machine Learning: Methods and Applications*. Berlin, Germany: Springer, 2012.

[28] H. Drucker, "Improving regressors using boosting techniques," in *Proc. Int. Conf. Mach. Learn.*, 1997, pp. 107–115.

[29] N. Noman and H. Iba, "Differential evolution for economic load dispatch problems," *Elect. Power Syst. Res.*, vol. 78, no. 8, pp. 1322–1331, 2008.

[30] F. Liang, C. Shen, W. Yu, and F. Wu, "Towards optimal power control via ensembling deep neural networks," *IEEE Trans. Commun.*, vol. 68, no. 3, pp. 1760–1776, Mar. 2020.

[31] M. Wang, H. B. Gooi, S. Chen, and S. Lu, "A mixed integer quadratic programming for dynamic economic dispatch with valve point effect," *IEEE Trans. Power Syst.*, vol. 29, no. 5, pp. 2097–2106, Sep. 2014.

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